Yet Another Video Summarization Tool

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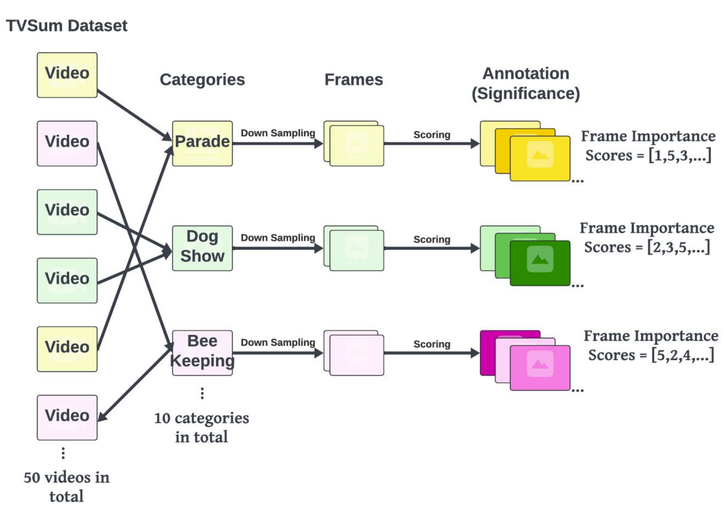
Introduction

The explosive rise of social media platforms such as TikTok has heralded a cultural shift towards concise, information-rich videos. Yet, the creation of these TikTok-style clips often demands expertise in video editing and access to high-performance hardware, presenting a high entry barrier for marginalized groups. To democratize video creation, we present Yet Another Video Summarization Tool (YAVST). Building on 2D CNN and transformer, YAVST allows the automatic synthesis of short, engaging videos from lengthy, naturally recorded footage

TVSum Dataset

Since YAVST is designed for general purpose use, we would like to train it on a dataset that is as diverse as possible. For this purpose, we chose TVSum[1] to be the training dataset as it contains well-annotated videos from multiple categories.

TVSum Examples



*Dataset is split into train (80%) and test (20%), stratified in the category label.

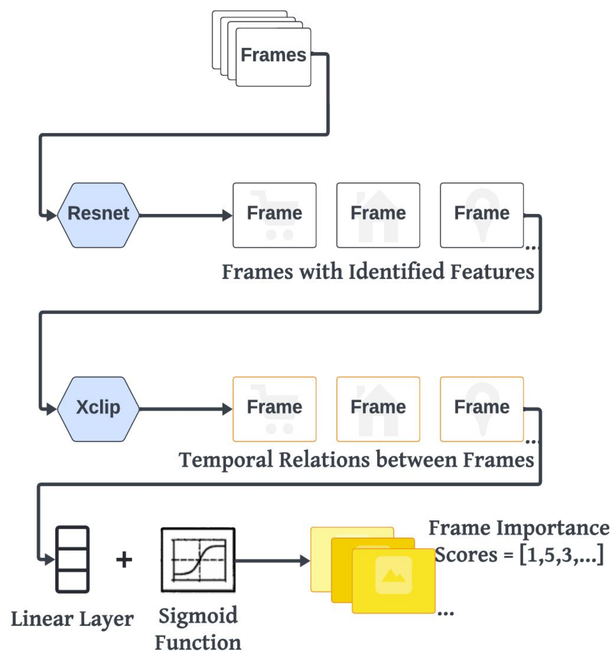
Methodology

Model Architecture

Resnet[2] is the SOTA 2D CNN model for feature extraction from each video frame

XClip[3] utilizes the SOTA cross-frame attention mechanism to gain temporal information of events occuring in a set of frames

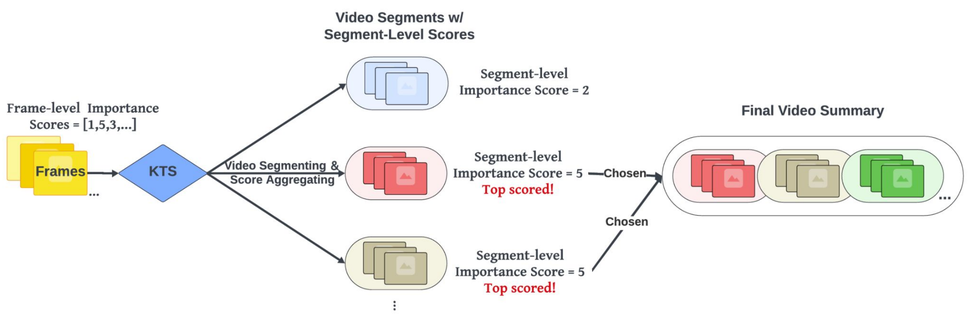
A linear layer with sigmoid function for computing importance



Parameters of Resnet were frozen during training

Video Summary Generation

- Kernel Temporal Segmentation (KTS)[4] was used to divide a video into segments.
- The frame importance score was aggregated to segment score, and top scoring segments form the final video output



Evaluation

A more robust evaluation over random[5] was used for video summarization methods to overcome shortcomings from small dataset and user-biased labeling.

models	F1	Number of Parameters
LSTM encoder-decoder	0.602	$\approx 8.5 \times 10^{10}$
adaptive rnn	0.467	$\approx 5.3 \times 10^{8}$
encoder-decoder with attention mechanism	0.639	$\approx 4 \times 10^{9}$
ours	TBD	TBD

Case Study

We employed YAVST to generate a video summary for a [type] video, here are some of the frames that YAVST deems (un)important

Video Category: Dog Show

important frames

unimportant frames

Conclusion

- We leveraged Resnet's feature extraction capability and XClip's cross-frame attention mechanism to build YAVST for general video summarization tasks
- Our performance analysis and case study showed that model training converged to a bad local minimum
- Further fine-tuning of training process is needed for accurate evaluation of YAVST

References

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[4] Potapov, D. et al; ECCV 2014: Computer Vision – ECCV 2014, 2014, pp. 540–555
[5] Apostolidis, E. et al. (2020) 'Performance over random'